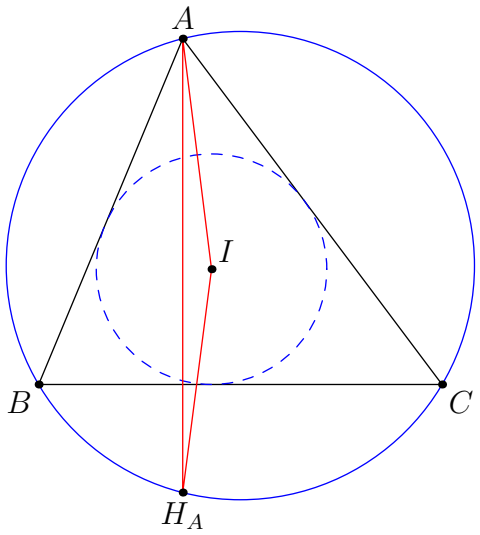
2022 SSMO Team Round Solutions

SMO Team

TR 1: In triangle ABC, circumcircle ω is drawn. Let I be the incenter of $\triangle ABC$ and H_A be the intersection of the A-altitude and ω. Given that $AB = 13$, $AC = 15$, and $BC = 14$, the area of triangle AIH_A can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{71}$

Solution:



We will use analytic geometry. Let $B = (0, 0)$, $A = (5, 12)$, and $C = (14, 0)$. We compute $I = (6, 4)$ and $H_A = \left(5, -\frac{15}{4}\right)$. From the shoelace theorem, the area of the triangle is

$$\frac{63}{8} \implies 63 + 8 = \boxed{71}.$$

TR 2: Consider a lineup of 8 marbles, each colored black or white. Define the period of a lineup to be the smallest positive integer k such that the lineup appears as a contiguous subsequence of an infinite repetition of a length-k lineup. An ordering is called good if its period is at most 6. The probability that a random lineup is good can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{89}$

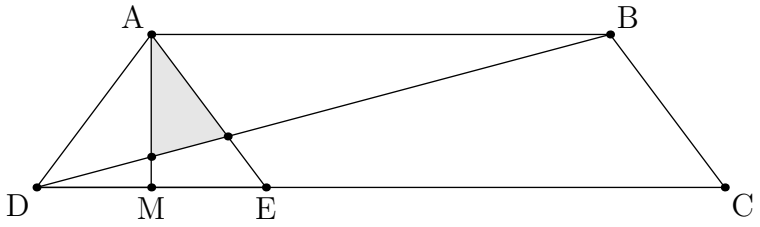
Solution: A lineup has period k exactly when the marbles k positions apart have the same color, and it does not have any smaller period. The first k marbles determine the entire lineup, so checking the 2^k possible choices for them gives the following counts of lineups with each smallest period.

Period	1	2	3	4	5	6
Count	2	2	6	12	30	48

Thus, the probability that a random lineup has period at most 6 is

$$\frac{2+2+6+12+30+48}{2^8} = \frac{100}{256} = \frac{25}{64} \implies 25+64 = \boxed{89}.$$

TR 3: Let $ABCD$ be an isosceles trapezoid such that $AB \parallel CD$. Let E be a point on CD such that $AB = CE$. Let M be the midpoint of DE. Denote G and F as the intersections of BD with AM and AE, respectively. Given that $DC = 36$, $AB = 24$, and $AD = 10$, the area $[AGF]$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.



Answer: $\boxed{69}$

Solution: We have $DM = ME = \frac{36-24}{2} = 6$, meaning $AM = \sqrt{10^2 - 6^2} = 8$. From $\triangle AGB \sim \triangle MGD$, we have

$$\frac{AG}{GM} = \frac{AB}{DM} = \frac{24}{6} = 4.$$

Since $AG + GM = AM = 8$, this means $AG = \frac{4}{4+1} \cdot 8 = \frac{32}{5}$. From $\triangle AFB \sim \triangle EFD$, we have

$$\frac{d(F, AB)}{d(F, DE)} = \frac{AB}{DE} = \frac{24}{12} = 2.$$

Since $d(F, AB) + d(F, DE) = AM = 8$, this means $d(F, AB) = \frac{2}{2+1} \cdot 8 = \frac{16}{3}$. So,

$$[AGF] = [ABD] - [AGD] - [AFB] = \frac{24 \cdot 8}{2} - \frac{\frac{32}{5} \cdot 6}{2} - \frac{24 \cdot \frac{16}{3}}{2} = \frac{64}{5} \implies 64+5 = \boxed{69}.$$

TR 4: Let $a_1 = 2$, $b_1 = 3$, $a_n = (a_{n-1})^2$, $b_n = (b_{n-1})^3$, and $c_n = a_n + b_n$. Find the last two digits of $c_1 + c_2 + \cdots + c_{2022}$.

Answer: $\boxed{56}$

Solution: From computation, we have

$$\begin{aligned}
a_1 &= 2 \\
a_2 &= 2^2 = 4 \\
a_3 &= 4^2 = 16 \\
a_4 &= 16^2 = 256 \equiv 56 \pmod{100} \\
a_5 &\equiv 56^2 \equiv 3136 \equiv 36 \pmod{100} \\
a_6 &\equiv 36^2 \equiv 1296 \equiv 96 \pmod{100} \\
a_7 &\equiv 96^2 \equiv 9216 \equiv 16 \pmod{100}
\end{aligned}$$

and

$$\begin{aligned}
b_1 &= 3\\
b_2 &= 3^3 = 27\\
b_3 &= 27^3 = 19683 \equiv 83 \pmod{100}\\
b_4 &\equiv 83^3 \equiv 571787 \equiv 87 \pmod{100}\\
b_5 &\equiv 87^3 \equiv 658503 \equiv 3 \pmod{100}.
\end{aligned}$$

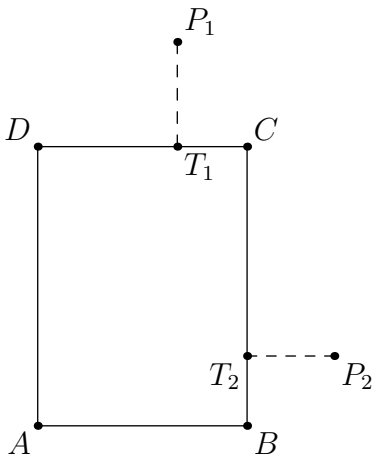
We have

$$\begin{aligned}
c_1 + \cdots + c_{2022} &= (a_1 + \cdots + a_{2022}) + (b_1 + \cdots + b_{2022})\\
&= [(2+4) + (16+56+36+96)\cdot 505]\\
&\quad + [(3+27+83+87)\cdot 505 + (3+27)]\\
&= [6 + 204\cdot 505] + [200\cdot 505 + 30]\\
&\equiv [6 + 4\cdot 5] + [0\cdot 5 + 30] \equiv \boxed{56} \pmod{100}.
\end{aligned}$$

TR 5: Consider the following rectangle $ABCD$ with $BC = 8$. Let T_1 and T_2 be the feet of the perpendiculars from P_1 and P_2 to CD and BC, respectively. Suppose

$$CD = CT_2, \quad 4T_2P_2 = 2DP_1 = AC, \quad [ADP_1C] = 33, \quad \text{and} \quad [ABP_2C] = 34.$$

Find $[P_1CP_2A]$.



Answer: $\boxed{45}$

Solution: Let $AB = x$. We have

$$AC = \sqrt{AB^2 + BC^2} = \sqrt{x^2 + 64} \implies T_2P_2 = \frac{AC}{4} = \frac{\sqrt{x^2+64}}{4}.$$

Now,

$$\begin{aligned}
[ABP_2C] &= [ABC] + [BCP_2]\\
&= \frac{AB\cdot BC}{2} + \frac{BC\cdot T_2P_2}{2} \implies\\
34 &= \frac{8\cdot x}{2} + \frac{8\cdot \frac{\sqrt{64+x^2}}{4}}{2} \implies\\
34 &= 4x + \sqrt{64+x^2} \implies x = 6.
\end{aligned}$$

Now,

$$\begin{aligned}[ADP_1C] &= [ADC] + [DCP_1] \\ &= \frac{AD \cdot DC}{2} + \frac{DC \cdot T_1P_1}{2} \implies \\ 33 &= \frac{6 \cdot 8}{2} + \frac{6 \cdot T_1P_1}{2} \implies T_1P_1 = 3.\end{aligned}$$

So,

$$DT_1 = \sqrt{DP_1^2 - T_1P_1^2} = \sqrt{\left(\frac{10}{2}\right)^2 - 3^2} = 4$$

and

$$T_2B = CB - CT_2 = CB - CD = 8 - 6 = 2.$$

We have

$$\begin{aligned}[P_1CP_2A] &= [ADP_1C] + [ABP_2C] - [ADP_1] - [ABP_2] \\ &= 33 + 34 - \frac{AD \cdot DT_1}{2} - \frac{AB \cdot T_2B}{2} \\ &= 33 + 34 - \frac{8 \cdot 4}{2} - \frac{6 \cdot 2}{2} = \boxed{45}.\end{aligned}$$

TR 6: For integers $a \geq 2$ and $x \geq 1$, let $P_{x,a}$ be the maximum fraction of elements in the set of the first x natural numbers that may be contained in a subset S such that if k is an element of S, then ak is not. For example, $P_{3,3} = \frac{2}{3}$, since we take the set $\{1, 2\}$. As x approaches infinity, $P_{x,a}$ approaches a value P_a. The value of $\prod_{a=2}^{100} P_a$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{103}$

Solution: We claim that $P_a = \frac{a}{a+1}$. Each chain k, ak, a^2k, ... of length L can contain at most $\left\lceil \frac{L}{2} \right\rceil$ elements of S, and choosing elements greedily from largest to smallest achieves this bound. This means we first take the top $\frac{a-1}{a}$ fraction, then skip the next $\frac{a-1}{a^2}$ fraction, then take the next $\frac{a-1}{a^3}$ fraction, and so on. For example, in the case $P_{64,2}$, this procedure selects $(33, 34, \ldots, 63, 64)$, then $(9, 10, \ldots, 15, 16)$, then $(3, 4)$, and finally 1. Thus we obtain

$$P_a = \frac{a-1}{a} + \frac{a-1}{a^3} + \frac{a-1}{a^5} + \ldots = \frac{\frac{a-1}{a}}{1 - \frac{1}{a^2}} = \frac{a}{a+1}.$$

So,

$$\prod_{a=2}^{100} P_a = \frac{2}{3} \cdot \frac{3}{4} \cdots \frac{100}{101} = \frac{2}{101} \implies 2 + 101 = \boxed{103}.$$

TR 7: Let $\cos(2A)$, $\cos(2B)$, and $\cos(2C)$ be the not necessarily distinct roots of a monic cubic f. Given that $f(1) = \frac{17}{30}$, the positive value of $\sin(A)\sin(B)\sin(C)$ can be expressed as $\frac{\sqrt{m}}{n}$, where m and n are positive integers and m is not divisible by the square of any prime. Find $m + n$.

Answer: $\boxed{315}$

Solution: Note that $\cos(2x) = 1 - 2\sin^2(x) \implies 1 - \cos(2x) = 2\sin^2(x)$. We have

$$\begin{aligned} f(1) &= (1 - \cos(2A))(1 - \cos(2B))(1 - \cos(2C)) \\ &= (2\sin^2(A))(2\sin^2(B))(2\sin^2(C)) \\ &= 8(\sin(A)\sin(B)\sin(C))^2. \end{aligned}$$

So,

$$\sin(A)\sin(B)\sin(C) = \sqrt{\frac{f(1)}{8}} = \sqrt{\frac{17}{240}} = \frac{\sqrt{255}}{60} \implies 255 + 60 = \boxed{315}.$$

TR 8: A frog is at 0 on a number line and wants to go to 9. On each turn, if the frog is at n, the frog hops to one of the numbers from n to 9, inclusive, with equal probability (staying in place counts as a hop). It is then teleported to the largest multiple of 3 that is less than or equal to the frog's position. The expected number of hops it takes for the frog to reach 9 can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{173}$

Solution: Let E_x be the expected number of moves it takes to reach 9 from x. We have

$$\begin{aligned} E_9 &= 0 \\ E_6 &= \frac{3}{4}E_6 + \frac{1}{4}E_9 + 1 \implies E_6 = 4 \\ E_3 &= \frac{3}{7}E_3 + \frac{3}{7}E_6 + \frac{1}{7}E_9 + 1 \implies E_3 = \frac{19}{4} \\ E_0 &= \frac{3}{10}E_0 + \frac{3}{10}E_3 + \frac{3}{10}E_6 + \frac{1}{10}E_9 + 1 \implies E_0 = \frac{145}{28} \implies 145 + 28 = \boxed{173}. \end{aligned}$$

TR 9: Real numbers a, b, x, y satisfy

$$\begin{aligned} a^2 + b^2 &= 1, \\ x^2 + y^2 &= 1, \quad \text{and} \\ abxy - \frac{1}{8} &= b^2y^2. \end{aligned}$$

Find the sum of all distinct values of $(a + b + x + y)^2$.

Answer: $\boxed{8}$

Solution: Let $(a, b) = (\cos\alpha, \sin\alpha)$ and $(x, y) = (\cos\beta, \sin\beta)$. We have

$$\begin{aligned} abxy - \frac{1}{8} &= b^2y^2 \\ by(ax - by) &= \frac{1}{8} \\ \sin\alpha\sin\beta(\cos\alpha\cos\beta - \sin\alpha\sin\beta) &= \frac{1}{8} \\ \left(\frac{\cos(\alpha - \beta) - \cos(\alpha + \beta)}{2}\right)\cos(\alpha + \beta) &= \frac{1}{8}. \end{aligned}$$

Let $t = \cos(\alpha+\beta)$ be fixed. The left hand side is maximized when $\cos(\alpha-\beta) = \pm 1$, depending on the sign of $\cos(\alpha+\beta)$. So,

$$(\cos(\alpha-\beta)-\cos(\alpha+\beta))\cos(\alpha+\beta) \le (\pm 1 - t)\,t = -\left(t \mp \frac{1}{2}\right)^2 + \frac{1}{4} \le \frac{1}{4}.$$

Since

$$(\cos(\alpha-\beta)-\cos(\alpha+\beta))\cos(\alpha+\beta) = \frac{1}{8},$$

we must have an equality case in the inequality above. If $t = \frac{1}{2}$, we have

$$\cos(\alpha+\beta) = \frac{1}{2} \quad \text{and} \quad \cos(\alpha-\beta) = 1,$$

giving $(\alpha,\beta) = \left(\frac{\pi}{6},\frac{\pi}{6}\right), \left(\frac{5\pi}{6},\frac{5\pi}{6}\right)$, meaning $a+b+x+y = \sqrt{3}\pm 1$. If $t = -\frac{1}{2}$, we have

$$\cos(\alpha+\beta) = -\frac{1}{2} \quad \text{and} \quad \cos(\alpha-\beta) = -1,$$

giving $(\alpha,\beta) = \left(\frac{5\pi}{6},\frac{11\pi}{6}\right), \left(\frac{7\pi}{6},\frac{\pi}{6}\right)$, meaning $a+b+x+y = 0$. The sum of all distinct values of $(a+b+x+y)^2$ is

$$(1+\sqrt{3})^2 + (1-\sqrt{3})^2 + 0^2 = \boxed{8}.$$

TR 10: Let p, q, r be the three roots of $x^3 - 2x^2 - 4$. Find the value

$$(p^3+qr)(q^3+pr)(r^3+pq).$$

Answer: $\boxed{400}$

Solution: From Vieta's formulas, we have

$$p+q+r = 2, \quad pq+pr+qr = 0 \quad \text{and} \quad pqr = 4.$$

So,

$$\begin{aligned} p^3+qr &= p^3 + (pq+pr+qr) - (pq+pr) \\ &= p^3 - pq - pr = p(p^2-(q+r)) \\ &= p(p^2-(2-p)) = p(p+2)(p-1) \\ &= p(-2-p)(1-p). \end{aligned}$$

Since p, q, r are the roots of $x^3 - 2x^2 - 4$, we have

$$(x-p)(x-q)(x-r) = x^3 - 2x^2 - 4.$$

So,

$$\begin{aligned} &(p^3+qr)(q^3+pr)(r^3+pq) \\ =&(pqr)\left((-2-p)(-2-q)(-2-r)\right)\left((1-p)(1-q)(1-r)\right) \\ =&(4)\left((-2)^3-2(-2)^2-4\right)\left(1^3-2(1^2)-4\right) \\ =&(4)(-20)(-5) = \boxed{400}. \end{aligned}$$

TR 11: Find the number of solutions to $a^3 + 36ab + 64 = 27b^3$, where a is an integer, b is a real number, and $|b| \leq 50$.

Answer: $\boxed{302}$

Solution: We factor

$$a^3 + 36ab + 64 - 27b^3 = (a - 3b + 4)(a^2 + 3ab - 4a + 9b^2 + 12b + 16).$$

So, either

$$a - 3b + 4 = 0 \quad \text{or} \quad a^2 + 3ab - 4a + 9b^2 + 12b + 16 = 0.$$

If the former, then $a = 3b - 4$. From $|b| \leq 50$, we have

$$3(-50) - 4 \leq a \leq 3(50) - 4 \implies -154 \leq a \leq 146,$$

giving 301 solutions. Otherwise, the latter must hold. Rewriting the equation as a quadratic in terms of a, we have

$$a^2 + (3b - 4)a + (9b^2 + 12b + 16) = 0.$$

The discriminant is

$$(3b - 4)^2 - 4(9b^2 + 12b + 16) = -3(3b + 4)^2 \leq 0,$$

meaning we must have $b = -\frac{4}{3} \implies a = 4$. Since $4 - 3\left(-\frac{4}{3}\right) + 4 \neq 0$, this is an additional solution. This gives a total of

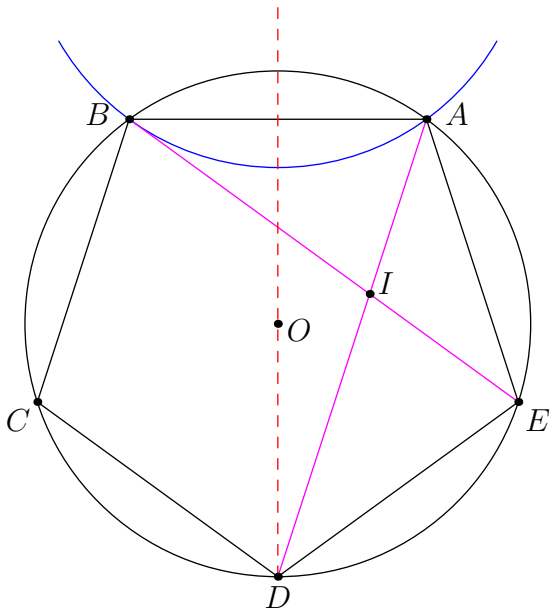
$$301 + 1 = \boxed{302}$$

solutions.

TR 12: Regular pentagon $ABCDE$ is inscribed in circle ω_1 centered at O with radius $5\sqrt{5}$. Circle ω_2 is the reflection of ω_1 across AB. Let I be the intersection of lines AD and BE, let P be an intersection of line DO and ω_2, and let line ℓ be the tangent to ω_2 at P. The sum of the possible distances from point I to line ℓ can be expressed as $m\sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m + n$.

Answer: $\boxed{20}$

Solution:



Position $ABCDE$ such that AB is horizontal. Note that DO is vertical, and it follows that both positions of ℓ are horizontal. Scaling down by a factor of $5\sqrt{5}$ to the unit circle, the y-coordinate of A is $\sin(54^\circ)$. So, the two positions of ℓ have y-coordinates $2\sin(54^\circ)-1$ and $2\sin(54^\circ)+1$. Now, we use complex numbers to find the y-coordinate of I. Let $\omega = e^{\frac{\pi i}{10}}$. Note that $A = \omega^3$, $B = \omega^7$, $D = \omega^{15}$, and $E = \omega^{19}$. We have

$$I = \frac{\omega^3\omega^{15}(\omega^7+\omega^{19}) - \omega^7\omega^{19}(\omega^3+\omega^{15})}{\omega^3\omega^{15} - \omega^7\omega^{19}} = \omega^7(1-\omega^4+\omega^8).$$

So, the y-coordinate of I is

$$\sin(7\cdot 18^\circ) - \sin(11\cdot 18^\circ) + \sin(15\cdot 18^\circ) = \sin(54^\circ) + \sin(18^\circ) - 1.$$

The sum of the two possible distances is

$$\begin{aligned} &((2\sin(54^\circ)-1) - (\sin(54^\circ)+\sin(18^\circ)-1)) \\ &+ ((2\sin(54^\circ)+1) - (\sin(54^\circ)+\sin(18^\circ)-1)) \\ &= 2(\sin(54^\circ) - \sin(18^\circ)) + 2 = 3. \end{aligned}$$

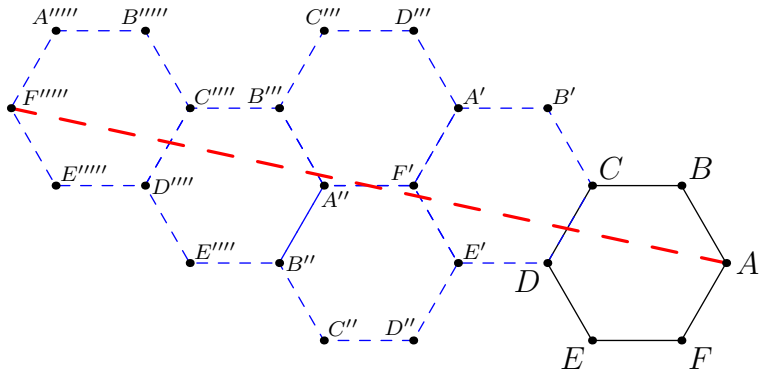
Scaling back up, we get

$$5\sqrt{5}\cdot 3 = 15\sqrt{5} \implies 15+5 = \boxed{20}.$$

TR 13: In regular hexagon $ABCDEF$ with side length 1, an electron starts at point A. When the electron hits an edge, it reflects off of it, with the angle of reflection equal to the angle of incidence. The electron first travels in a straight line to a point on edge CD. The electron bounces off of a total of 5 edges before hitting a vertex. The electron stops and its total distance traveled is measured. The shortest possible distance the electron could have traveled can be expressed as $\sqrt{m}$. Find m.

Answer: $\boxed{67}$

Solution:



We repeatedly reflect the hexagon to achieve a ring of hexagons. Our answer is the shortest path from A to a vertex X on the ring such that AX passes through 5 edges of the ring of hexagons. Shown above is the shortest possible path, which involves passing through CD, EF, AF, AB, and CD, which has a length of

$$\sqrt{8^2 + \left(\sqrt{3}\right)^2} = \sqrt{67} \implies \boxed{67}.$$

TR 14: On a hot summer day, three little piggies decide to play with water balloons. The three piggies travel to a 200-floor parking garage each armed with exactly one water balloon. The game works as follows:

1. If a piggy drops a water balloon from any floor of the building, it will either break, or it will survive the fall.
2. If the water balloon breaks, then any greater fall would have broken it as well.
3. If the water balloon survives, then it would have survived any lesser fall.
4. Every water balloon is identical and interchangeable.

The goal for the piggies is to find the lowest floor that will break a water balloon. Assuming they play optimally, what is the minimum number of tries in which they are guaranteed to find the lowest balloon-breaking floor?

Answer: $\boxed{11}$

Solution: Let $F(k,t)$ be the maximum number of floors that can be handled with k balloons and t drops. If the first drop is from floor $F(k-1,t-1)+1$, then a break leaves $k-1$ balloons and $t-1$ drops for the floors below it, while a survival leaves k balloons and $t-1$ drops for the floors above it. So,

$$F(k,t) = 1 + F(k-1,t-1) + F(k,t-1).$$

Note that $F(1,t) = t$, since with one balloon we have to start from floor 1 and iterate upwards. Now,

$$\begin{aligned} F(2,t) &= 1 + F(1,t-1) + F(2,t-1) \\ &= t + F(2,t-1) \\ &= t + (t-1) + \cdots + 1 = \frac{t(t+1)}{2}. \end{aligned}$$

So,

$$\begin{aligned} F(3,t) &= 1 + F(2,t-1) + F(3,t-1) \\ &= 1 + \frac{(t-1)(t)}{2} + F(3,t-1) \\ &= t + \frac{(t-1)(t)}{2} + \frac{(t-2)(t-1)}{2} + \dots \\ &= t + \frac{(t-1)(t)(t+1)}{6}. \end{aligned}$$

The minimum value of t that satisfies

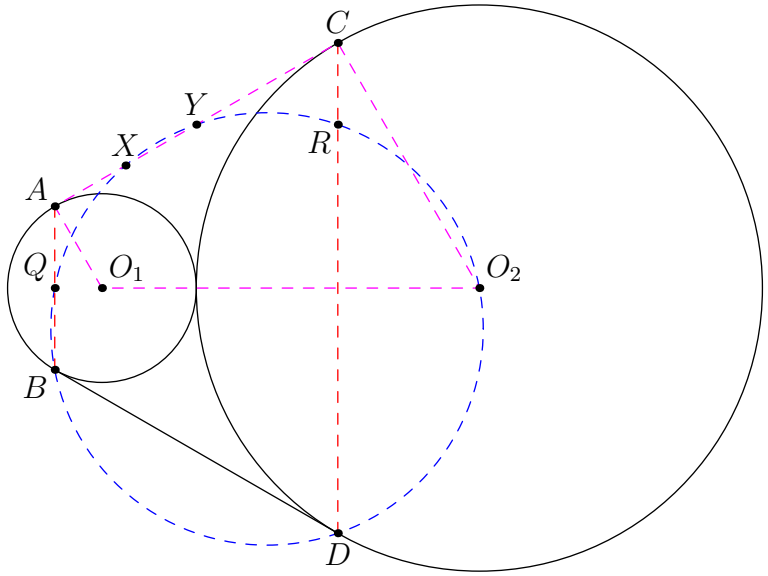
$$F(3,t) = t + \frac{(t-1)(t)(t+1)}{6} \geq 200$$

is $t = \boxed{11}$, with $F(3,t-1) = 175$ and $F(3,t) = 231$.

TR 15: Consider two externally tangent circles ω_1 and ω_2 with centers O_1 and O_2. Suppose that ω_1 and ω_2 have radii of 1 and 3 respectively. There exist points A, B on ω_1 and points C, D on ω_2 such that AC and BD are the external tangents of ω_1 and ω_2. The circumcircle of $\triangle BO_2D$ intersects AC at two points X and Y such that $AX < AY$. The length of CX can be expressed as $\frac{a\sqrt{b}}{c}$, where a, b, and c are positive integers, a and c are relatively prime, and b is not divisible by the square of any prime. Find $a+b+c$.

Answer: $\boxed{8}$

Solution:



Firstly, we have

$$AC = \sqrt{(O_1O_2)^2 - (CO_2 - AO_1)^2} = \sqrt{4^2 - (3-1)^2} = 2\sqrt{3}.$$

It follows that

$$AB = \sqrt{3} \qquad \text{and} \qquad CD = 3\sqrt{3}.$$

Now, let Q and R be the intersections of AB and CD with the circumcircle of $\triangle BO_2D$, respectively. Since BQO_2D is cyclic, we have

$$\angle BQO_2 = 180^\circ - \angle BDO_2 = 180^\circ - 90^\circ = 90^\circ.$$

So, Q is the midpoint of AB, giving $AQ = \frac{\sqrt{3}}{2}$. Since BO_2 is a diameter, the foot of its midpoint on CD is the midpoint of DR, which gives $CR = \frac{\sqrt{3}}{2}$. From the power of a point, we have

$$(AX)(AY) = (AX)(AX + XY) = (AQ)(AB) = \frac{\sqrt{3}}{2} \cdot \sqrt{3}$$
$$(CY)(CX) = (CY)(XY + YC) = (CR)(CD) = \frac{\sqrt{3}}{2} \cdot 3\sqrt{3}$$
$$AX + XY + YC = AC = 2\sqrt{3}.$$

Solving, we have $AX = \frac{\sqrt{3}}{2}$, $XY = \frac{\sqrt{3}}{2}$, and $YC = \sqrt{3}$, meaning

$$CX = XY + YC = \frac{\sqrt{3}}{2} + \sqrt{3} = \frac{3\sqrt{3}}{2} \implies 3 + 3 + 2 = \boxed{8}.$$